is accompanied by his abbreviated name. Example: *Pithecellobium cinereum* Benth.

**Multiple authors**: When two or more authors are associated with a valid publication of name, their names should be noted with the help of Latin word *et* or &.

Example: *Delphinium viscosum* Hook. f. *et* Thomson.

**Standard form of author's abbreviations has to be followed.**

| Author | Standard form of Abbreviation |
|---|---|
| Linnaeus | L. |
| G.Bentham | Benth. |
| William Hooker | Hook. |
| Robert Brown | R.Br. |
| J.P.Lamarck | Lamk. |
| A.P.de Candolle | DC. |
| Wallich | Wall. |
| Alphonse de Candolle | A. DC. |

## 5.5 Taxonomic Aids

Taxonomic aids are the tools for the taxonomic study. Some techniques, procedures and stored information that are useful in identification and classification of organisms are called **taxonomical aids.** They are required in almost all branches of biological studies for their proper identification and for finding their relationship with others. Some of the taxonomical aids are keys, flora, revisions, monograph, catalogues, herbarium, botanical gardens etc.

### Keys

Taxonomic keys are the tools for the identification of unfamiliar plants. These keys are based on characters which are stable and reliable. The most common type of key is a dichotomous key. It consists of a sequence of two contrasting statements. A pair of contrasting statements is known as **couplet.** Each statement is known as **lead.** The plant is correctly identified with keys by narrowing down the characters found in plant.

Example:

1. a) Flowers cream-coloured; fruiting calyx enclosing the berry .....*Physalis*
   b) Flowers white or violet; fruiting calyx not enclosing the berry: ....*2*
2. a) Corolla rotate; fruit a berry .....*Solanum*
   b) Corolla funnel-form or salver-form; fruit a capsule: ....*3*
3. a) Radical leaves present; flowers in racemes; fruits without prickles ...*Nicotiana*
   b) Radical leaves absent; flowers solitary; fruits with prickles .....*Datura*

Another type of key for identification is the **Polyclave** or **Multi-entry key**. It consists of a list of numerous character states. The user selects all states that match the specimen. Polyclave keys are implemented by a computer algorithm.

## 5.6 Botanical Gardens

In true sense all gardens are not botanical gardens. Botanical gardens are centres for collection of plants in their various stages of living. Gardens existed for growing ornamental plants for aesthetic value, religious and status reasons. The famous "hanging gardens" of Babylon in Mesopotamia is an example. For the purpose of science and education the first garden was maintained by **Theophrastus** in his public lecture hall at Athens.

First modern botanical garden was established by **Luca Ghini** (1490-1556) a professor of Botany at Pisa, Italy in 1544.

Botanical garden contains special plant collections such as cacti, succulent, green house, shade house, tropical, alpine and exotic plants. Worldwide there are about 1800 botanical gardens and arboreta.

**Role of Botanical Garden:** Botanical Gardens play the following important roles.

1. Gardens with aesthetic value which attract



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